

Two plants, one criterion, thirty-two weeks

The plan re-cut around the real contribution: not a controller for engines, but a criterion for when preview information is worth acquiring — tested on two systems with deliberately different dynamics.

PLANTS	AIRCRAFT	TEAM	BANKED BY
Engine (built) + battery	Discussed, not built	5, working as one unit	Week 18

1 · Orientation

What changed, and why the plan had to be re-cut

The project's contribution moved. It used to be *"a predictive supervisory controller for engines"* — which scores badly on originality, because Mercedes-Benz ships predictive powertrain control on trucks and reinforcement learning for engine control is an active published field with national-lab work behind it.

It is now **a criterion for when preview information is worth acquiring at all**. That question is open, nobody owns it, and it has a real user: any control engineer deciding whether to invest in map data, extra sensors or a communication link needs to know whether the investment will pay before building it.

THE EVIDENCE THAT JUSTIFIED THE CHANGE

Sweeping the turbine thermal time constant while holding the preview horizon at 30 s, with identical policies and identical actuators, the preview advantage moved from **7.6 points to 49.9 points** — a 7× swing driven purely by a dimensionless ratio. Preview value is not a property of engines. It is a property of H/τ .

DECISION	VERDICT	REASONING
Keep the engine plant	CORE	Built, validated against factory figures, and it anchors the work in a real vehicle you own.
Add an EV battery plant	CORE	Different physics, different actuators, and — critically — it sweeps H/τ across two orders of magnitude. See §3.
Aircraft engine	DISCUSS	The most interesting case and the one you cannot validate. Reasoned extrapolation in Chapter 7, explicitly labelled untested.
Heavy truck	DROPPED	Near-identical physics to the car with more mass. Real weeks, almost no new information, and it is where the commercial prior art already sits.
Pareto front over preferences	DEMOTED	Was a headline result; now a stretch goal. The criterion story does not need the full trade-off surface, and something had to give.

What the thesis now answers

RQ1 · PER PLANT What is preview worth? Fuel or energy, and accumulated damage, against lookahead horizon. Run on both plants. Already answered on the engine with hand-written policies.	RQ2 · THE HEADLINE Does H/τ predict it? Across both plants and a swept range of horizons and time constants, does one dimensionless ratio predict how much preview is worth? This is the contribution.	RQ3 · METHODS Does pessimism help? How much apparent gain is surrogate error being exploited. Cheap to run, and it is the machine-learning contribution rather than the control one.
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Note the ordering. RQ1 is now a *supporting* result rather than the headline — it is the measurement RQ2 generalises. If an examiner asks what is new, the answer is RQ2, and RQ1 is the evidence base underneath it.

Why the battery, specifically

A criterion cannot be tested on one point. The engine alone can only reach $H/\tau \approx 0.6$ — its turbine constant is around 50 s and map preview gives you at most 30 s. Everything below that is a smaller ratio; there is no way to go above it.

A battery pack changes that completely, because route knowledge is available at *many* horizons:

PREVIEW AVAILABLE	HORIZON H	H/T (T ≈ 19 MIN)	REGIME
Immediate road ahead	30 s	0.03	Preview should be worthless
Next manoeuvre	2 min	0.11	Marginal
Next segment of route	10 min	0.53	Comparable to the engine
Remaining journey	30 min	1.60	Beyond anything the engine reaches
Full planned route	60 min	3.20	Should saturate — test this

THIS IS THE ARGUMENT TO GIVE YOUR SUPERVISOR

The battery is not "a second example." It is the instrument that makes the criterion *testable*, because it spans two orders of magnitude in the very ratio the criterion is about. One plant gives you an anecdote; two plants with overlapping but differently-ranged H/τ gives you a curve — and a prediction you can check.

The falsifiable claim: **preview value should collapse onto a single curve when plotted against H/τ , regardless of which plant produced the point.** If the two plants' curves do not overlap, the criterion is wrong or incomplete, and saying so is still a result.

4 · The plan

Six stages

All five members on one stage at a time, three lanes per session, leadership rotating. Each stage has a gate you can check and a fallback you take rather than overrunning.

WEEKS	LEAD	STAGE	WHAT GETS BUILT	GATE · AND FALLBACK IF MISSED
1–4	all	Sprint 0 Nobody specialises	Everyone runs the existing plant, thermal network, environment and premise check. Each member teaches one concept to the other four. Interfaces frozen. Leads assigned.	Gate: every member independently reproduces one existing result and writes a one-page note on it. No fallback — do not skip this.
5–11	1	Engine hardening	Turbocharger flow model closing the 16 % power error. Re-validation against factory figures. Logging campaign begins in week 5 regardless of code state.	Gate: peak power within 5 % of 387 PS; validation table still passes. Fallback: fixed boost ceiling as a function of speed instead of a compressor map. Costs realism, saves two weeks.
9–16	1	Vehicle campaign overlaps — weather-bound	≈8 h of logging on the Supra. Priority channels first. Taif escarpment runs for the sustained-climb cycle. Calibrate the port heat-loss constant.	Gate: ≥6 h clean logs; plant torque and fuel within 10 % on covered points. Fallback: validate against published dyno data, state the substitution in Ch 3.
11–18	2	Battery plant + first criterion result THE PIVOT	Equivalent-circuit + thermal + Arrhenius aging model (§7). Route-based drive cycles. Then run the horizon sweep on both plants with hand-written policies.	Gate: battery plant hits its validation targets, and you have a first H/τ curve from both plants. Fallback: none needed — this stage is the insurance.
18–23	3	Unified environment + baselines	One environment both plants plug into via <code>predict()</code> . Reactive baseline controllers for each. MPC oracle with perfect preview.	Gate: neutral action ≈ 0 reward on both plants; MPC beats each baseline measurably. Fallback: drop MPC, keep reactive baselines. Costs the "is learning necessary" chapter.
21–28	4	Surrogates + the agent	Ensembles for both plants with the disagreement signal. Constrained SAC with PID Lagrange multipliers. Trained on both.	Gate: beats the reactive baseline on both plants, five seeds, at equal or better constraint satisfaction. Fallback: fixed-penalty single-objective agent. RQ2 survives on hand policies.

WEEKS	LEAD	STAGE	WHAT GETS BUILT	GATE · AND FALLBACK IF MISSED
27–31	5	The criterion experiment	Full sweep: horizons × time constants × both plants × 5 seeds, with the learned agent. The collapse plot. Pessimism ablation for RQ3.	Gate: all figures regenerate from committed result files with one command. Fallback: report the hand-policy version from week 18 as the primary result.
26–32	all	Writing lane 3 drafting since week 5	Each lead writes their chapter. Aircraft extrapolation section. Mock individual defences.	Gate: full draft to supervisor week 30, leaving two weeks of real revision.

THE SINGLE MOST IMPORTANT LINE IN THIS TABLE

Stage 4 ends at **week 18** with a complete answer to the headline question, produced by hand-written policies on both plants. No neural network is involved.

That means from week 18 onward you are *improving* a result you already have rather than racing to produce one. Everything after that stage is upside. Protect week 18 above every other date in this document.

5 · Checklist

Milestones, in order

Print this. Tick items only when the stated evidence exists — not when the work "feels done."

WK	✓	MILESTONE – AND THE EVIDENCE THAT IT IS ACTUALLY DONE
1	☐	All five run <code>plant.py</code> , <code>thermal.py</code> , <code>engine_env.py</code> , <code>check_premise.py</code> on their own machines
2	☐	Five internal teach-backs delivered — knock, Wiebe/MBT, lambda, thermal network, constrained MDP
3	☐	Three interface contracts frozen and written down; integration test runs on <code>main</code>
3	☐	First exploratory OBD log taken — you now know what the Supra actually exposes
4	☐	Leads assigned; each has written their own research question and agreed it with the supervisor
7	☐	Turbo flow model in; peak power error down from 16 % to under 5 %
10	☐	Engine validation table re-passes after the turbo change (regression test, not eyeball)
13	☐	Battery plant produces plausible voltage sag, temperature rise and capacity fade
14	☐	Battery aging rate roughly doubles per +10 °C — the Arrhenius sanity check
15	☐	Route-based drive cycles built; preview available at 30 s / 2 / 10 / 30 / 60 min
16	☐	≥6 h of clean vehicle logs captured; port heat-loss constant calibrated
18	☐	H/τ curve produced from BOTH plants with hand-written policies. The thesis now has a result.
20	☐	Unified environment: both plants behind one <code>predict()</code> ; neutral action ≈ 0 on each

WK	✓	MILESTONE — AND THE EVIDENCE THAT IT IS ACTUALLY DONE
23	☐	MPC oracle completes all cycles on both plants and beats both reactive baselines
25	☐	Surrogate ensembles trained; disagreement verified to rise on a held-out region
28	☐	Constrained agent beats reactive baseline on both plants across five seeds
30	☐	All figures regenerate from committed results with one command
30	☐	Full draft with supervisor
31	☐	Mock individual defences run — each member interrogated by the other four

6 • Forks

Decisions you must make on time

Each of these has a deadline. Making them late is what turns a slip into a crisis.

BY WEEK	THE DECISION	DECIDE IT THIS WAY
3	Which logging tool	Take the exploratory log first. If the generic PIDs give you spark advance at ≥ 5 Hz, that may be enough for calibration and you save €109. If knock retard is absent and you want it, buy the monitor licence now — not in week 12 when the weather turns.
10	Compressor map or boost ceiling	If a credible compressor map is not running by week 9, take the speed-dependent ceiling. It fixes the same error less elegantly and buys two weeks you will want later.
13	Battery fidelity	If the equivalent-circuit model is fighting you, drop the RC pair and use pure ohmic resistance. Aging and thermal behaviour — the parts the criterion actually needs — barely change.
18	Go / no-go on the learned agent HARD GATE	If the H/τ curve is not in hand from both plants by week 18, stop expanding and consolidate . Drop the surrogate ensemble and RQ3, train directly on the plants, and protect the criterion result. This is the one date that is not negotiable.
23	MPC oracle in or out	If MPC is not completing cycles by week 23, cut it. Keep the reactive baselines and answer "why not MPC" analytically in the discussion instead of empirically.
27	Learned agent or hand policies for the headline figure	Whichever is more complete on week 27 becomes the primary result; the other becomes a supporting section. Decide by looking at the data, not by sunk cost.

7 • Build spec

The battery plant

Everything needed to write it. Target: a 65 kWh, 400 V liquid-cooled pack, roughly a mid-size EV. Same `predict()` contract as the engine so it drops into the existing environment.

Three layers, mirroring the engine plant

Electrical – equivalent circuit

$$V_{\text{terminal}} = \text{OCV}(\text{SoC}) - I \cdot R_0 - V_{\text{RC}}, \quad dV_{\text{RC}}/dt = (I \cdot R_1 - V_{\text{RC}}) / (R_1 C_1)$$

$$\dot{Q}_{\text{gen}} = I^2 \cdot R_0 + V_{\text{RC}} \cdot I$$

R_0 and R_1 both rise as temperature falls – that coupling is what makes cold packs limit power.

Thermal – lumped capacitance, same form as the engine

$$C_{\text{pack}} \cdot dT/dt = \dot{Q}_{\text{gen}} - UA(\text{coolant flow}) \cdot (T - T_{\text{coolant}})$$

$$C_{\text{pack}} \approx 450 \text{ kJ/K} \cdot UA \approx 200\text{--}800 \text{ W/K} \cdot \text{therefore } \tau \approx 9\text{--}38 \text{ min}$$

Aging – Arrhenius calendar term plus a throughput term

$$dQ_{\text{loss}}/dt = A \cdot \exp(-E_a / R \cdot T) \cdot (1 + k \cdot |I| / C_{\text{rate,ref}})$$

$E_a \approx 50 \text{ kJ/mol}$ reproduces the classic "aging rate doubles per +10 °C" (verified: $\times 1.92$)

Actions the agent gets

ACTION	RANGE	PREVIEW RELEVANCE
Coolant flow / chiller duty	0 ... 1	Pre-cool before a known fast-charge stop or a sustained climb
Discharge power limit	−30 ... +10 %	Spread thermal load across a known demanding segment
Charge-rate limit	−40 ... 0 %	Trade charging time against aging when arrival time has slack
Cabin/pack thermal split	0 ... 1	Where cooling capacity goes when both want it

Validation targets — the equivalent of the engine's table

BEHAVIOUR	EXPECTED
Pack voltage sag at 300 A	4–8 %
Temperature rise over a 150 kW DC fast charge	10–20 °C
Capacity fade at 1000 full cycles, 25 °C	10–20 %
Aging rate multiplier per +10 °C	$\approx 2\times$
Pack thermal time constant	10–40 min

BEHAVIOUR	EXPECTED
Usable power at $-10\text{ }^{\circ}\text{C}$ vs $25\text{ }^{\circ}\text{C}$	40–70 %

ACCEPT THE PLANT ONLY WHEN

Every row above lands in range *without being fitted to it*, exactly as with the engine. And one extra check specific to this plant: **cold-temperature power limiting must emerge from the resistance-temperature coupling**, not be imposed as a rule. If you have to hand-write "if cold, limit power," the electrical layer is wrong and the criterion study will inherit the error.

8 · The measurement

Computing H/τ , and the plot that is the thesis

The ratio is only meaningful if τ is measured the same way on both plants. Fix the procedure now and write it into Chapter 4, because an examiner will ask whether you chose τ to make the curves line up.

measuring tau – identical procedure on both plants

§8

```
# tau is the time constant of the DOMINANT CONSTRAINED state - the one whose
# limit the controller is actually defending. Engine: turbine housing.
# Battery: pack temperature. Measure it, do not read it off the parameters.

def measure_tau(plant, step_input, settle_s=4000):
    """Open-loop step response. tau = time to reach 63.2% of the final change.
    Run with the controller DISABLED - you are characterising the plant,
    not the closed loop."""
    y = plant.step_response(step_input, duration=settle_s)
    y0, yf = y[0], y[-1]
    target = y0 + 0.632 * (yf - y0)
    return np.argmax(y >= target) * plant.dt

# The headline figure. Every point is one (plant, horizon, tau) triple.
# If the criterion holds, points from BOTH plants fall on one curve.
for plant in [engine, battery]:
    tau = measure_tau(plant, step_input=plant.nominal_load_step)
    for H in plant.available_horizons:
        edge = preview_advantage(plant, H) # predictive - reactive, in points
        record(x=H / tau, y=edge, plant=plant.name)
```

WHAT THE FIGURE HAS TO SHOW, AND WHAT WOULD FALSIFY IT

Supports the criterion: engine points and battery points fall on one curve when plotted against H/τ , despite different physics, actuators and units. Preview value low at small H/τ , rising through a band, saturating above $H/\tau \approx 1-2$.

Falsifies it: the two plants produce clearly separated curves. That would mean H/τ is not sufficient and at least one more dimensionless group is needed — most likely something capturing actuator authority, or how much of the disturbance you can actually pre-absorb. *Report that honestly if it happens.* A criterion shown to be incomplete, with the missing dimension identified, is a better thesis than a criterion asserted without test.

9 · Templates

Copy these on day one

experiments/configs/example.yaml — one file per run, never hardcode

T1

```
name: horizon_sweep_battery_h600
plant:
  kind: battery          # battery | engine
  preset: pack_65kwh
  overrides: {ua_coolant: 400.0}
cycle:
  kind: route
  file: cycles/jeddah_taif_return.csv
  t_ambient_k: 315.0
controller:
  kind: hand_predictive  # hand_neutral | hand_reactive | hand_predictive | sac | mpc
  preview_horizon_s: 600
evaluation:
  evaluated_on: plant    # plant | surrogate - ALWAYS state this
  seeds: [0, 1, 2, 3, 4]
  deterministic: true
```

results/*.json — the schema every experiment writes

T2

```
{
  "config":      {...},          # the resolved config, not a path to it
  "git_sha":     "a1b2c3d",
  "seed":        0,
  "plant":       "battery",
  "evaluated_on": "plant",       # REQUIRED - refuse to write without it
  "tau_measured_s": 1104.0,      # measured, not assumed
  "horizon_s":   600.0,
  "h_over_tau":  0.543,
  "metrics": {
    "energy": ..., "energy_baseline": ...,
    "damage": ..., "damage_baseline": ...,
    "constraint_violation_rate": ...,
    "peak_constrained_state": ...
  },
  "trace_path":  "traces/....parquet"
}
```

```
git pull && pytest -q           # integration test FIRST, before any new work
python -m experiments.smoke_all  # both plants, neutral action, reward ~ 0
python -m experiments.regen_figures # figures still build from committed results?

# then, in the standup: each member shows ONE PLOT from last week.
# not a status update. plots surface problems that words hide.
```

10 · Notes

Things that will cost you a week if you learn them late

On the engine stage

- **Take the first OBD log in week 3, not week 9.** You need to know what your car exposes before you plan around it. If EGT turns out to be modelled rather than measured — likely on a B58 — that changes what the calibration chapter can claim, and you want to know in month one.
- **Pin the existing behaviour with a regression test before touching the plant.** The turbo model changes airflow, which touches knock, which touches everything. Without a test you will not notice you broke the validated behaviour until week 20.
- **Jeddah in September is an asset.** The Taif escarpment at 40 °C+ is the grade-climb cycle. Most groups have to simulate that condition; you can drive to it. Log it early while the weather holds.

On the battery stage

- **Do not model individual cells.** A single lumped pack is enough for a criterion study and cell-level modelling will eat a month for no gain in what you are measuring.
- **The aging law is the load-bearing part, not the electrical model.** If time is short, simplify the circuit and keep the Arrhenius term careful — it is what makes preview valuable in this plant.
- **Route cycles need real elevation.** Reuse the GPS traces from the engine campaign; the same Jeddah–Taif route drives both plants, which is a nice consistency for the cross-plant comparison.

On the criterion experiment

- **Measure τ , never read it off parameters.** The measured closed-loop constant differs from C/UA , and using the analytic value invites the accusation that you tuned it to make the curves align.
- **Fix the τ measurement procedure before you produce any points,** and write it down. Changing it later invalidates everything measured before the change.
- **Plot preview advantage in points, not percent of baseline.** Percentages compress at the top — that is what made the convexity experiment read as non-monotonic when it was partly a measurement artifact.

On working as five

- **Timer on the driver rotation.** 25 minutes, rotate mid-function if needed. Without it, two people write everything within three weeks.
- **Lane 3 from week 5.** Someone is always writing. This is the least glamorous line in the plan and the most likely to save it.
- **Tag the repository at every result.** When someone asks in week 30 how a figure was made, the answer should be a tag, not archaeology.

11 · Triage

If you fall behind, cut in this order

Decided in advance and in cold blood, so that a bad month becomes a smaller thesis rather than a panic.

CUT	WHAT GOES	WHAT YOU LOSE · WHAT SURVIVES
1st	Preference conditioning / Pareto front	Lose the trade-off surface figure. RQ1, RQ2 and RQ3 all survive intact.
2nd	MPC oracle	Lose the empirical answer to "is learning necessary." Answer it analytically instead and flag it as a limitation.
3rd	Surrogate ensembles → train on plants directly	Lose RQ3 entirely. Slower training, no loss of validity. Already measured: the engine runs a full episode in ~42 s.
4th	The learned agent	Report the hand-policy criterion result as primary. The thesis still stands — this is what week 18 protects.
5th	The battery plant	Fall back to the single-plant thesis with the H/τ sweep done by varying the engine's thermal mass. Weaker generality claim, still a complete thesis.
never	The vehicle campaign	It is what makes this a real-vehicle project rather than a simulation exercise, and it is cheap. Even 4 hours of logs earns the chapter.

12 · Risk

Updated for two plants

RISK	LIKELY	IMPACT	MITIGATION
The two plants' curves do not collapse onto one	Medium	Medium	This is a result, not a failure. Identify the missing dimensionless group — most likely actuator authority relative to disturbance size — and report the criterion as necessary but insufficient. Prepare this framing in advance so it does not feel like a rescue.
Battery plant eats more than 7 weeks	Medium	High	Fidelity ladder is pre-decided in §6: drop the RC pair, then drop temperature-dependent resistance, then accept a fixed-resistance pack. Aging and thermal must survive; everything else is negotiable.

RISK	LIKELY	IMPACT	MITIGATION
Scope creep back toward aircraft	Medium	High	It is genuinely the most interesting case, which is exactly why someone will want to build it around week 20. Do not. It cannot be validated, and an unvalidated third plant weakens the two you can defend.
Sequential critical path — one slip costs everyone	High	Medium	Inherent to working as one unit. The gates are the countermeasure: miss one, take the fallback, keep moving. And week 18 banks the headline result early.
Reward hacking goes unnoticed	High	Low	Likely but cheap to catch. Log every reward component separately; read action traces at every checkpoint, not just returns. Budget for finding two or three.
Vehicle exposes neither knock retard nor EGT	Medium	Low	Found out in week 3 by design. Calibrate against published data and state it. The plant does not depend on the campaign to function.

The plan in one sentence. Harden the engine, build the battery, get the H/τ curve from both with hand-written policies by week 18, then spend the remaining fourteen weeks making that result better rather than making it exist.

The three dates that matter. Week 3, first OBD log — because it decides your data plan. Week 18, criterion result banked — because it decides whether you have a thesis. Week 30, draft to supervisor — because two weeks of revision is the difference between a pass and a good grade.